

1.1.Mid



## Daffodil International University

Department of Software Engineering

Faculty of Science & Information Technology

Mid Semester Examination Spring 2026

Course Code: SE113; Course Title: Introduction to Software Engineering

Sections & Teachers: IM (47-A), RM (47-B, 47-C, 47-D, 47-E), NML (47-F), FF (47-G)

Duration: 1:30 Hours

Marks: 25

**Answer ALL Questions**

*[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]*

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                 |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|
| 1. | <p>A software company is hired to develop a National Online Library Management System.</p> <p>Project characteristics:</p> <ul style="list-style-type: none"> <li>The system will serve thousands of users</li> <li>Requires requirement analysis, design, implementation, testing, and maintenance</li> <li>Some modules (search, login, reporting) can be developed separately</li> <li>The government demands documentation and formal reviews</li> <li>The team wants to understand which development process and models are appropriate before starting</li> </ul> <p>Answer the following questions based on the above scenario.</p> |             | CLO1<br>Level 2 |
|    | a) Explain the importance of Software Engineering practices for this project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | [Marks – 4] |                 |
|    | b) Describe the phases of SDLC and explain how each phase would contribute to this project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [Marks – 5] |                 |
|    | c) Differentiate between Waterfall and Spiral models using examples from the scenario.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [Marks – 4] |                 |
|    | <p>d) The management considers using the V model.</p> <p>Explain how this model would work for this project and describe two advantages and two disadvantages of this model.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [Marks – 5] |                 |
| 2. | <p>A startup is developing a Food Delivery Mobile App.</p> <p>Project characteristics:</p> <ul style="list-style-type: none"> <li>Requirements change every week</li> <li>Customers frequently request new features</li> <li>The team is small (6 developers)</li> <li>Fast release and continuous improvement are required</li> <li>Work items include both new features and bug fixes</li> </ul> <p>As a software engineer involved in the above project, select and apply an appropriate modern methodology among Scrum, XP, or Kanban.</p>                                                                                             |             | CLO2<br>Level 3 |
|    | a) Select the most suitable methodology and demonstrate reasoning behind your selection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [Marks – 3] |                 |
|    | b) Demonstrate how the software engineers will organize the workflow (form/graph/roles/events/board/iterations).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [Marks – 4] |                 |